Theoretical Knowledge

Basic Incident Response

Basic incident response refers to the set of procedures and actions taken by security teams to detect, investigate, contain, and recover from cybersecurity incidents. An effective incident response process helps organizations minimize damage, reduce recovery time, and prevent similar incidents from occurring in the future.

The incident response process is typically structured into several phases, commonly defined in the **NIST SP 800-61 Incident Handling Guide**. The first phase is **Preparation**, which involves establishing the tools, policies, and procedures required for responding to incidents. This includes developing incident response playbooks, configuring monitoring tools, training SOC analysts, and establishing communication protocols.

The second phase is **Identification**, where potential security incidents are detected and analyzed. Alerts generated by security tools are reviewed by SOC analysts to determine whether they represent real threats or false positives. During this phase, analysts collect relevant information such as log data, network traffic details, and system activity to understand the nature of the incident.

Once an incident has been confirmed, the next phase is **Containment**. The goal of containment is to limit the spread and impact of the attack. This may involve isolating infected systems from the network, disabling compromised user accounts, or blocking malicious IP addresses through firewall rules.

After containment, the **Eradication** phase focuses on removing the root cause of the incident. This may include deleting malware, patching vulnerabilities, removing malicious user accounts, or cleaning compromised systems.

The next phase is **Recovery**, where systems and services are restored to normal operation. This may involve restoring data from backups, rebuilding compromised systems, and verifying that systems are functioning securely before reconnecting them to the network.

The final phase is **Lessons Learned**, which involves reviewing the incident and identifying improvements that can strengthen the organization's security posture. Security teams analyze what happened, how the incident was detected, and how response procedures can be improved to prevent future incidents.

In addition to these phases, incident response procedures include several important activities such as system isolation, evidence preservation, and forensic analysis. Analysts must ensure that digital evidence is collected carefully and preserved in a way that maintains its integrity. Techniques such as memory dumps, disk imaging, and file hashing are commonly used to collect and verify digital evidence.

Communication is also a critical component of incident response. Security teams must communicate effectively with management, technical teams, and other stakeholders throughout

the incident response process. Structured reporting methods such as Situation Reports (SITREPs) help provide clear and concise updates about ongoing incidents.

The objective of learning basic incident response is to equip analysts with the knowledge and procedures required to respond effectively to security incidents. A well-structured response process allows organizations to detect threats quickly, limit damage, and recover from attacks efficiently.

To develop incident response skills, analysts can study the NIST SP 800-61 Incident Handling Guide, which provides detailed guidance on incident management. The SANS Incident Handler's Handbook offers practical response templates and real-world examples of incident handling procedures. Additionally, cybersecurity training platforms such as Let's Defend provide simulated incident response scenarios that allow analysts to practice detection, investigation, and response techniques in controlled environments.